

# A $\text{Spin}^c$ Projection-Dolbeault Derivation of Three Standard Model Generations in PSLT

Bo-Yu Chen<sup>1</sup>

<sup>1</sup>*Independent Researcher*

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We formulate the Projection Spectral Layer Theory (PSLT) as a first-principles projection-spectral derivation of the three-generation multiplicity of the Standard Model chiral family representation, conditional on the PSLT two-center projection data. The compact projection surface is  $C = \mathbb{CP}^1$ , the two projection branch points define the divisor  $R = p_+ + p_-$ , and the generation line bundle is  $L_R = \mathcal{O}(R) \simeq \mathcal{O}(2)$ . The internal fermion operator is specified to be the  $\text{Spin}^c$  Dolbeault–Dirac operator

$$\mathcal{D}_R = \sqrt{2} \left( \bar{\partial}_{L_R} + \bar{\partial}_{L_R}^\dagger \right)$$

acting on  $\Lambda^{0,*} T^* C \otimes L_R$ . Hodge theory gives

$$\text{Ker } \mathcal{D}_R^+ \cong H^0(C, L_R), \quad \text{Ker } \mathcal{D}_R^- \cong H^1(C, L_R).$$

For  $C = \mathbb{CP}^1$  and  $L_R = \mathcal{O}(2)$ , Riemann–Roch and Serre duality give

$$h^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3, \quad h^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0.$$

Provided the SM-charged chiral carrier is exactly this  $\text{Spin}^c$  generation sector and all other SM-charged internal sectors are absent from the kernel or fully paired and gapped, the low-energy four-dimensional chiral spectrum contains exactly three Standard Model families and no unpaired light SM-charged exotics. The same two-center geometry supplies the conformal factor  $\Omega$  from a variational Poisson constraint, the scalar spectral operator  $[-\nabla^2 + V_{\text{eff}}]\psi_n = \omega_n^2 \psi_n$ , and the scalar spectral data used in an effective formation/visibility probability law

$$P_N(t) = \frac{B_N g_N^{\text{chiral}} (1 - e^{-\Gamma_N^{\text{eff}} t})}{\sum_{K=1}^3 B_K g_K^{\text{chiral}} (1 - e^{-\Gamma_K^{\text{eff}} t})}.$$

The scalarization weights  $\mu_{nN}$  select a hard-locked mode only when the locking gap exists, in which case  $\psi_N^{\text{lock}} := \psi_{n(N;D)}$ ,  $\omega_N := \omega_{n(N;D)}$ , and  $\Gamma_N^{\text{eff}} = \Gamma_N$ ; otherwise  $\Gamma_N^{\text{eff}} = \Gamma_N^{\text{soft}}$ . Thus the generation count is a  $\text{Spin}^c$  index-theoretic consequence of the two-center projection divisor, while  $(g_N^{\text{chiral}}, \Gamma_N^{\text{eff}}, B_N)$  describe degeneracy, kinetics, and visible-sector overlap of the three protected modes.

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## I. CORE STRUCTURE

The construction has three logically distinct layers:

1. the *SM charge layer*, where one minimal Standard Model family is fixed by gauge invariance, Yukawa invariance, and anomaly cancellation;
2. the *internal fermion layer*, where the  $\text{Spin}^c$  Dolbeault–Dirac operator on the projection surface supplies the chiral zero-mode multiplicity;
3. the *formation and visibility layer*, where the two-center conformal geometry supplies scalar spectral data for effective rates and visible-sector overlaps.

The data needed for the  $\text{Spin}^c$  generation-multiplicity theorem are only

$$\boxed{\mathfrak{D}_{\text{gen}} = (C, R, L_R, \mathcal{D}_R).} \quad (1)$$

They fix  $\text{Ker } \mathcal{D}_R^+$ , hence the number of protected chiral generations. The effective formation/visibility layer uses additional interface data

$$\boxed{\mathfrak{D}_{\text{int}}(D) = (\pi_D, \mathfrak{j}_D, w_D, \mathcal{J}_{\text{vis}}).} \quad (2)$$

These interface data assign scalar profiles, rates, and visible overlaps to the protected modes; they are not inputs to the index-theoretic statement  $N_{\text{gen}} = 3$ . The core Hilbert-space statement is

$$\boxed{\mathcal{H}_{\text{chiral}}^{4\text{D}} = \mathcal{R}_{\text{SM}} \otimes \text{Ker } \mathcal{D}_R^+ = \mathcal{R}_{\text{SM}} \otimes H^0(\mathbb{CP}^1, \mathcal{O}(p_+ + p_-)).} \quad (3)$$

Since  $p_+ + p_-$  has degree two,

$$\mathcal{O}(p_+ + p_-) \simeq \mathcal{O}(2), \quad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3. \quad (4)$$

The low-energy chiral spectrum is therefore

$$\boxed{\mathcal{H}_{\text{chiral}}^{4\text{D}} = \mathcal{R}_{\text{SM}}^{(1)} \oplus \mathcal{R}_{\text{SM}}^{(2)} \oplus \mathcal{R}_{\text{SM}}^{(3)}} \quad (5)$$

The family count is the dimension of the positive internal  $\text{Spin}^c$  fermion zero-mode space.

### A. Notation conventions

Throughout the paper,  $D$  denotes the two-center separation parameter. Dirac-type operators are written with script letters, such as  $\mathcal{D}_R$ ,  $\mathcal{D}_X$ ,  $\mathcal{D}_{\text{vec}}$ , and  $\mathcal{D}_{M_4}^{\text{SM}}$ . The Bergman projector onto

$H^0(C, L_R)$  is  $P_R$ , the collective-coordinate projector is  $P_N^{\text{cc}}$ , the one-dimensional Toeplitz-mode projector is  $\Pi_N = |u_N\rangle\langle u_N|$ , and the observed layer probability is  $P_N(t; D, \eta)$ . The protected zero-mode degeneracy is denoted  $g_N^{\text{chiral}}$ ; any unadorned degeneracy symbol is avoided in the chiral family count.

## II. PARENT PROJECTION DATA

Let  $M_4$  be four-dimensional spacetime and let the compact projection surface be

$$C \simeq \mathbb{CP}^1. \quad (6)$$

The two projection branch points define the divisor

$$R = p_+ + p_-, \quad L_R = \mathcal{O}(R) \simeq \mathcal{O}(2). \quad (7)$$

The parent action is organized as

$$S_{\text{FP}} = S_{\text{grav}} + S_{\text{gauge}} + S_{\text{Higgs}} + S_{\text{fermion}} + S_{\Omega}^{\text{cov}} + S_{\text{proj}} + S_{\text{vis}}. \quad (8)$$

The projection sector fixes the topological class of  $L_R$ , not a pointwise equality between a smooth curvature form and a delta-current. The robust condition is the cohomological constraint

$$\boxed{\left[ \frac{i}{2\pi} F_{A_R} \right] = \text{PD}(R) = [p_+] + [p_-] \in H^2(C, \mathbb{Z}), \quad R = p_+ + p_-} \quad (9)$$

Equivalently,

$$c_1(L_R) = \text{PD}(R), \quad \deg L_R = 2. \quad (10)$$

When an action-like expression is useful, it should be read as a topological constraint or as a regulated current limit:

$$S_{\text{proj}}^{\text{top}} = \int_C \lambda_R \left[ \frac{i}{2\pi} F_{A_R} - J_R^{(\varepsilon)} \right], \quad (11)$$

where  $J_R^{(\varepsilon)}$  is any smooth representative of  $\text{PD}(R)$  that tends, as a current, to  $\delta_{p_+}^{(2)} + \delta_{p_-}^{(2)}$ . The first-principles input is Eq. (9); Eq. (11) is only a regulated implementation of the same topological class.

### A. SM-charged fermion sector

The SM-charged internal fermion carrier is assumed to be exactly

$$\mathcal{E}_{\text{gen}} = \mathcal{R}_{\text{SM}} \otimes \mathcal{S}_C, \quad \mathcal{S}_C = \Lambda^{0,*} T^* C \otimes L_R. \quad (12)$$

Let

$$X = M_4 \times C. \quad (13)$$

The generation-sector product Dirac operator is

$$\boxed{\mathcal{D}_X = \mathcal{D}_{M_4}^{\text{SM}} \otimes 1 + \gamma_5 \otimes \mathcal{D}_R, \quad \Gamma_X = \gamma_5 \otimes \Gamma_C.} \quad (14)$$

Here  $\mathcal{D}_{M_4}^{\text{SM}}$  includes the four-dimensional SM gauge connection on  $\mathcal{R}_{\text{SM}}$ ,  $\Gamma_C$  is the internal  $\text{Spin}^c$  chirality, and  $\Gamma_X$  is the product chirality. The generation-sector fermion action is

$$\boxed{S_{\text{fermion}}^{\text{gen}} = \int_{M_4 \times C} \text{dvol}_{M_4} \text{dvol}_C \bar{\Psi} i \mathcal{D}_X \Psi.} \quad (15)$$

The internal mode expansion is

$$\Psi(x, y) = \sum_{\alpha} \psi_{\alpha}^{+}(x) u_{\alpha}^{+}(y) + \sum_{\beta} \psi_{\beta}^{-}(x) u_{\beta}^{-}(y) + \sum_{\lambda > 0} \Psi_{\lambda}(x) \chi_{\lambda}(y), \quad (16)$$

where

$$\mathcal{D}_R u_{\alpha}^{+} = 0, \quad u_{\alpha}^{+} \in \text{Ker } \mathcal{D}_R^{+}, \quad \mathcal{D}_R u_{\beta}^{-} = 0, \quad u_{\beta}^{-} \in \text{Ker } \mathcal{D}_R^{-}. \quad (17)$$

After a fixed product-chirality projection, internal positive and negative zero modes produce opposite four-dimensional Weyl chiralities. We choose the product chirality so that  $\text{Ker } \mathcal{D}_R^{+}$  carries the left-handed SM family representation  $\mathcal{R}_{\text{SM}}$ ;  $\text{Ker } \mathcal{D}_R^{-}$  would supply the opposite four-dimensional chirality in the same generation sector. Thus the chain is explicit:

$$\text{internal } \text{Spin}^c \text{ zero mode} \implies \text{massless 4D Weyl fermion} \implies \text{SM chiral generation.} \quad (18)$$

All other SM-charged internal sectors are required to be vectorlike and genuinely gapped:

$$\mathcal{E}_{\text{fermion}}^{\text{SM}} = \mathcal{E}_{\text{gen}} \oplus \mathcal{E}_{\text{vec}}. \quad (19)$$

The condition is stronger than  $\text{ind } \mathcal{D}_{\text{vec}} = 0$ . It requires either

$$\text{Ker } \mathcal{D}_{\text{vec}} = 0, \quad (20)$$

or

$$\begin{aligned} \dim \text{Ker } \mathcal{D}_{\text{vec}}^{+} &= \dim \text{Ker } \mathcal{D}_{\text{vec}}^{-}, \\ M_{\text{vec}} : \text{Ker } \mathcal{D}_{\text{vec}}^{+} &\longrightarrow \text{Ker } \mathcal{D}_{\text{vec}}^{-}, \quad M_{\text{vec}} \text{ is full rank and gauge invariant.} \end{aligned} \quad (21)$$

Index zero only excludes extra net chirality. Equation (20) or Eq. (21) is the additional low-energy condition that excludes unpaired massless SM-charged vectorlike exotics. SM-neutral sectors may be present, but they do not carry generation quantum numbers:

$$\mathcal{E}_{\text{fermion}} = \mathcal{E}_{\text{gen}} \oplus \mathcal{E}_{\text{vec}} \oplus \mathcal{E}_{\text{neutral}}. \quad (22)$$

This is the exclusivity condition needed for an exact no-fourth-generation theorem. Without it, an additional SM-charged internal bundle with nonzero index could produce extra chiral matter.

### III. ONE STANDARD MODEL FAMILY

The local anomaly equations use the Standard Model Lie algebra. When a global group and charge lattice are needed, we take

$$G_{\text{SM}} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6}. \quad (23)$$

This global choice fixes the conventional hypercharge unit used below; without it, the local equations determine hypercharge only up to an overall  $U(1)_Y$  normalization. For one left-handed Weyl family, take

$$Q : (3, 2)_{Y_Q}, \quad u^c : (\bar{3}, 1)_{Y_u}, \quad d^c : (\bar{3}, 1)_{Y_d}, \quad L : (1, 2)_{Y_L}, \quad e^c : (1, 1)_{Y_e}, \quad (24)$$

with Higgs

$$H : (1, 2)_h. \quad (25)$$

Set  $Y_Q = q$ . Gauge invariance of the Yukawa couplings gives

$$QH u^c : \quad q + h + Y_u = 0, \quad QH^\dagger d^c : \quad q - h + Y_d = 0, \quad (26)$$

$$LH^\dagger e^c : \quad Y_L - h + Y_e = 0. \quad (27)$$

Hence

$$Y_u = -q - h, \quad Y_d = -q + h, \quad Y_e = h - Y_L. \quad (28)$$

The  $SU(3)^2 U(1)$  anomaly gives

$$2Y_Q + Y_u + Y_d = 0, \quad (29)$$

which is automatic after Eq. (28). The pure color anomaly also cancels:

$$\mathcal{A}_{SU(3)^3} = 2A(3) + A(\bar{3}) + A(\bar{3}) = 2A(3) - A(3) - A(3) = 0, \quad (30)$$

where the factor of two comes from the two  $SU(2)_L$  components of  $Q$ , and  $A(\bar{3}) = -A(3)$ . The  $SU(2)^2 U(1)$  anomaly gives

$$3Y_Q + Y_L = 0, \quad Y_L = -3q. \quad (31)$$

The mixed gravitational anomaly requires

$$6Y_Q + 3Y_u + 3Y_d + 2Y_L + Y_e = 0. \quad (32)$$

Substitution gives

$$6q + 3(-q - h) + 3(-q + h) + 2(-3q) + (h + 3q) = 0, \quad (33)$$

so

$$h = 3q. \quad (34)$$

Therefore

$$Y_u = -4q, \quad Y_d = 2q, \quad Y_L = -3q, \quad Y_e = 6q, \quad Y_H = 3q. \quad (35)$$

The  $U(1)^3$  anomaly cancels:

$$6q^3 + 3(-4q)^3 + 3(2q)^3 + 2(-3q)^3 + (6q)^3 = 0. \quad (36)$$

There is no Witten global  $SU(2)$  anomaly. One family contains

$$3 \text{ color copies of } Q + 1 \text{ lepton doublet} = 4 \quad (37)$$

left-handed  $SU(2)_L$  Weyl doublets, which is even. For three families the number is 12, still even. With  $q = 1/6$ ,

$$\boxed{(Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H) = \left(\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1, \frac{1}{2}\right)}. \quad (38)$$

A sterile  $\nu^c : (1, 1)_0$  may be added without changing these anomaly equations.

**Proposition 1.** *Within the minimal one-family chiral field content above, and assuming the Yukawa couplings  $QH u^c$ ,  $QH^\dagger d^c$ , and  $LH^\dagger e^c$ , the local anomaly equations fix the hypercharge pattern up to the overall normalization of  $U(1)_Y$ . The choice in Eq. (23) fixes the conventional unit  $q = 1/6$ .*

#### IV. SPIN<sup>c</sup> DOLBEAULT–DIRAC ZERO MODES

The internal fermion operator is not the ordinary spin Dirac operator. It is the Spin<sup>c</sup> Dolbeault–Dirac operator

$$\boxed{\mathcal{D}_R = \sqrt{2} \left( \bar{\partial}_{L_R} + \bar{\partial}_{L_R}^\dagger \right)} \quad (39)$$

acting on

$$\mathcal{S}_C = \Lambda^{0,*} T^* C \otimes L_R. \quad (40)$$

*Remark 1* (BPS-compatible origin of the Dolbeault representative). In a BPS or worldline-supersymmetric realization of this protected generation sector, massless states are represented by  $Q$ -cohomology classes, with

$$Q^2 = 0, \quad H = \{Q, Q^\dagger\}. \quad (41)$$

For the holomorphic carrier

$$\mathcal{H}_{\text{int}} = \Omega^{0,*}(C, L_R), \quad (42)$$

locality, first-order gauge covariance, and compatibility with the complex structure select a Cauchy–Riemann type representative

$$Q = \bar{\partial}_{L_R} \quad (43)$$

up to holomorphic bundle isomorphism and gauge choice. The corresponding Hermitian Dirac representative is therefore

$$\mathcal{D}_R = \sqrt{2}(Q + Q^\dagger) = \sqrt{2}(\bar{\partial}_{L_R} + \bar{\partial}_{L_R}^\dagger). \quad (44)$$

This remark is only a BPS-compatible motivation for the protected PSLT sector used below; the theorem remains conditional on the two-center projection data, the holomorphic carrier  $\Omega^{0,*}(C, L_R)$ , and the absence of additional flux, superpotential, Ext-complex, or torsion deformations that would replace the Dolbeault differential by a different protected complex.

We use the canonical  $\text{Spin}^c$  structure on the complex curve  $C$ , whose untwisted spinor bundle is  $\Lambda^{0,*}T^*C$ . The line bundle  $L_R$  is an auxiliary twisting bundle for the Dolbeault–Dirac operator, not the  $\text{Spin}^c$  determinant line itself. The relevant index is therefore the Dolbeault index

$$\text{ind } \mathcal{D}_R = \chi(C, L_R) = h^0(C, L_R) - h^1(C, L_R). \quad (45)$$

Its internal chirality decomposition is

$$\mathcal{S}_C^+ = \Lambda^{0,0}T^*C \otimes L_R, \quad \mathcal{S}_C^- = \Lambda^{0,1}T^*C \otimes L_R. \quad (46)$$

By Hodge theory,

$$\boxed{\text{Ker } \mathcal{D}_R^+ \cong H^0(C, L_R), \quad \text{Ker } \mathcal{D}_R^- \cong H^1(C, L_R).} \quad (47)$$

This is the step that converts a holomorphic-section count into an internal fermion zero-mode theorem.



*Remark 2.* The  $\text{Spin}^c$  choice is essential for the stated bundle  $L_R = \mathcal{O}(2)$ . For an ordinary spin Dirac operator on  $\mathbb{CP}^1$ , the spin bundle contributes  $K_C^{1/2} = \mathcal{O}(-1)$ . Twisting an ordinary spin Dirac operator only by  $\mathcal{O}(2)$  would count

$$H^0(\mathbb{CP}^1, K_C^{1/2} \otimes \mathcal{O}(2)) = H^0(\mathbb{CP}^1, \mathcal{O}(1)),$$

which is two-dimensional. Equivalently, an ordinary spin-Dirac presentation would need the twist

$$E = K_C^{-1/2} \otimes L_R = \mathcal{O}(1) \otimes \mathcal{O}(2) = \mathcal{O}(3)$$

so that  $K_C^{1/2} \otimes E = L_R$ .

## V. THREE-GENERATION THEOREM

**Theorem 1** (PSLT  $\text{Spin}^c$  three-generation theorem). *Let  $C = \mathbb{CP}^1$ ,  $R = p_+ + p_-$ , and  $L_R = \mathcal{O}(R) \simeq \mathcal{O}(2)$ . Let the SM-charged generation carrier be*

$$\mathcal{R}_{\text{SM}} \otimes (\Lambda^{0,*} T^* C \otimes L_R),$$

*with internal operator  $\mathcal{D}_R$  as in Eq. (39). Assume the product Dirac action is Eq. (15), the product chirality is Eq. (14), every other SM-charged internal sector satisfies the full vectorlike gap condition of Eq. (20) or Eq. (21), and all remaining chiral sectors are SM-neutral. Then the low-energy four-dimensional chiral SM spectrum contains exactly three generations:*

$$N_{\text{gen}} = \dim \text{Ker } \mathcal{D}_R^+ - \dim \text{Ker } \mathcal{D}_R^- = 3. \quad (48)$$

*Moreover, since  $\text{Ker } \mathcal{D}_R^- = 0$ , the positive internal zero-mode space itself is three-dimensional.*

*Proof.* Riemann–Roch for a line bundle  $L$  on a compact Riemann surface gives

$$h^0(C, L) - h^1(C, L) = \deg L + 1 - g(C). \quad (49)$$

For  $C = \mathbb{CP}^1$ ,  $g(C) = 0$ . Because  $R = p_+ + p_-$ ,  $\deg L_R = 2$ . Therefore

$$h^0(C, L_R) - h^1(C, L_R) = 3. \quad (50)$$

Serre duality gives

$$h^1(C, L_R) = h^0(C, K_C \otimes L_R^{-1}). \quad (51)$$

On  $\mathbb{CP}^1$ ,

$$K_C = \mathcal{O}(-2), \quad L_R^{-1} = \mathcal{O}(-2), \quad (52)$$

so

$$K_C \otimes L_R^{-1} = \mathcal{O}(-4). \quad (53)$$

A negative-degree line bundle on  $\mathbb{CP}^1$  has no nonzero holomorphic sections. Thus

$$h^1(C, L_R) = 0, \quad h^0(C, L_R) = 3. \quad (54)$$

Using Eq. (47),

$$\dim \text{Ker } \mathcal{D}_R^+ = 3, \quad \dim \text{Ker } \mathcal{D}_R^- = 0. \quad (55)$$

The product Dirac reduction in Eqs. (14)–(17) turns these internal zero modes into massless four-dimensional Weyl fermions. The vectorlike gap condition prevents additional SM-charged chiral generations and also removes unpaired massless SM-charged vectorlike exotics.  $\square$

In an affine coordinate  $\zeta$  on  $\mathbb{CP}^1$ ,

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)) = \text{span}\{1, \zeta, \zeta^2\}. \quad (56)$$

This span is only a vector space until a physical basis is chosen. The family count is basis-independent; the labels  $N = 1, 2, 3$  require a canonical Hermitian operator.

## VI. CANONICAL GENERATION BASIS

The two branch points alone determine the divisor and the family count, but a physical basis also needs the normalization of the affine coordinate on  $C$ . Placing  $p_+$  at 0 and  $p_-$  at  $\infty$  leaves the residual freedom  $\zeta_R \mapsto a\zeta_R$ . In PSLT this freedom is fixed by the two-center projection map from the spatial chart to the compact projection surface.

### A. Projection map and coordinate normalization

Let  $(\rho, z_{\text{ax}}, \varphi)$  be cylindrical coordinates on  $\mathbb{R}^3$ , with the ordered centers

$$x_{\pm} : \quad \rho = 0, \quad z_{\text{ax}} = \pm D/2. \quad (57)$$

With regulator  $\varepsilon > 0$ , define

$$r_{\pm} = \sqrt{\rho^2 + (z_{\text{ax}} \mp D/2)^2 + \varepsilon^2}. \quad (58)$$

The physical affine coordinate is

$$\boxed{\zeta_R(\rho, z_{\text{ax}}, \varphi; D, \varepsilon) = \left(\frac{r_+}{r_-}\right)^{1/2} e^{i\varphi}.} \quad (59)$$

Strictly, Eq. (59) is a cylindrical-coordinate representative on  $\mathbb{R}^3 \setminus A_{\text{ax}}$ , where

$$A_{\text{ax}} := \{\rho = 0\}, \quad (60)$$

because the angular coordinate  $\varphi$  is undefined on the axis. The axial set has zero measure with respect to  $d\mu_{\text{cyl}} = \rho d\rho dz_{\text{ax}} d\varphi$ , so the pullback construction used in the visibility integrals may be read in an  $L^2$ -almost-everywhere sense. Equivalently, one may smooth the axial phase and take the regularized limit. With this convention, we write the projection map as a map from the spatial chart:

$$\boxed{\pi_D : \mathbb{R}^3 \longrightarrow C \simeq \mathbb{CP}^1, \quad \pi_D(\rho, z_{\text{ax}}, \varphi) = [1 : \zeta_R(\rho, z_{\text{ax}}, \varphi; D, \varepsilon)].} \quad (61)$$

In the collapsed-core limit  $\varepsilon \rightarrow 0$ ,

$$\pi_D(x_+) = p_+ = [1 : 0], \quad \pi_D(x_-) = p_- = [0 : 1]. \quad (62)$$

For finite  $\varepsilon$ , the source cores are mapped to regulated neighborhoods of the branch points. The normalization is fixed by the midplane condition

$$z_{\text{ax}} = 0 \implies |\zeta_R| = 1, \quad (63)$$

and the residual  $U(1)$  phase is fixed by the origin of the axial angle  $\varphi$ . Thus the earlier scale freedom  $\zeta_R \mapsto a\zeta_R$  is removed by the physical projection geometry rather than by an arbitrary Gram–Schmidt convention.

## B. Toeplitz basis

Let  $P_R$  be the Bergman projector from  $L^2(C, L_R)$  onto  $H^0(C, L_R)$ , using the Fubini–Study metric on  $C = \mathbb{CP}^1$  and the induced Hermitian metric on  $L_R$ . Using the coordinate  $\zeta_R$  fixed by Eq. (59), the associated axial moment-map function is

$$\mu_R(\zeta_R, \bar{\zeta}_R) = \frac{|\zeta_R|^2 - 1}{|\zeta_R|^2 + 1}. \quad (64)$$

Define the finite-dimensional Toeplitz operator

$$\boxed{\mathcal{T}_R = P_R M_{\mu_R} P_R} \quad (65)$$

on  $H^0(C, L_R)$ , where  $M_{\mu_R}$  denotes multiplication by  $\mu_R$ . The generation basis is defined by

$$\mathcal{T}_R u_N = \tau_N u_N, \quad u_N \in H^0(C, L_R), \quad N = 1, 2, 3. \quad (66)$$

For the symmetric two-center placement and the standard Fubini–Study/Bergman inner product, this basis is equivalent, up to normalization and ordering, to the spin-one monomial basis

$$1, \quad \zeta, \quad \zeta^2. \quad (67)$$

The Toeplitz definition is the physical statement; the monomial basis is only a convenient coordinate representation.

## VII. TWO-CENTER CONFORMAL GEOMETRY

The two-center conformal factor is obtained in the static sector. The covariant parent term  $S_\Omega^{\text{cov}}$  in Eq. (8) is not specified by the spatial expression below; rather, under the static two-center ansatz it reduces to the three-dimensional variational energy functional

$$E_\Omega = \frac{\kappa}{2} \int_{\mathbb{R}^3} d^3x |\nabla \Omega|^2 - 4\pi\kappa \int_{\mathbb{R}^3} d^3x \sigma(x)(\Omega - 1), \quad (68)$$

with

$$\sigma(x) = a [\rho_\varepsilon(x - x_+) + \rho_\varepsilon(x - x_-)]. \quad (69)$$

Variation gives

$$\delta E_\Omega = -\kappa \int_{\mathbb{R}^3} d^3x (\nabla^2 \Omega + 4\pi\sigma) \delta\Omega, \quad (70)$$

hence

$$\boxed{\nabla^2 \Omega = -4\pi\sigma.} \quad (71)$$

Let  $(\rho, z_{\text{ax}}, \varphi)$  be cylindrical coordinates on  $\mathbb{R}^3$ , with centers at

$$x_\pm : \quad \rho = 0, \quad z_{\text{ax}} = \pm D/2. \quad (72)$$

For

$$\rho_\varepsilon(r) = \frac{3\varepsilon^2}{4\pi(r^2 + \varepsilon^2)^{5/2}}, \quad \int_{\mathbb{R}^3} d^3x \rho_\varepsilon = 1, \quad (73)$$

the asymptotically flat solution is

$$\boxed{\Omega(\rho, z_{\text{ax}}; D) = 1 + a \left( \frac{1}{r_+} + \frac{1}{r_-} \right), \quad r_\pm = \sqrt{\rho^2 + (z_{\text{ax}} \mp D/2)^2 + \varepsilon^2}.} \quad (74)$$

### VIII. GEOMETRY-TO-SPECTRUM MAP

We use the mostly-plus convention

$$\eta_{\mu\nu} = \text{diag}(-, +, +, +), \quad \square_\eta = -\partial_t^2 + \nabla^2, \quad (75)$$

and the curvature convention

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}. \quad (76)$$

For the static conformally flat metric

$$g_{\mu\nu} = \Omega^2(x) \eta_{\mu\nu}, \quad (77)$$

this gives

$$R = -6\Omega^{-3}\nabla^2\Omega. \quad (78)$$

Consider

$$(\square_g - m_0^2 - \xi R)\Phi = 0, \quad \Phi(t, x) = e^{-i\omega t} \Omega^{-1}(x) \psi(x). \quad (79)$$

Using

$$\left( \square_g - \frac{1}{6}R \right) (\Omega^{-1}\psi) = \Omega^{-3} \square_\eta \psi, \quad (80)$$

one obtains

$$[-\nabla^2 + m_0^2 \Omega^2 + (1 - 6\xi) \Omega^{-1} \nabla^2 \Omega] \psi = \omega^2 \psi. \quad (81)$$

Define

$$V_{\text{eff}} = m_0^2 \Omega^2 + (1 - 6\xi) \Omega^{-1} \nabla^2 \Omega. \quad (82)$$

Then

$$\boxed{[-\nabla^2 + V_{\text{eff}}] \psi_n = \omega_n^2 \psi_n.} \quad (83)$$

This scalar spectral problem governs formation barriers and visible overlap profiles. It does not determine the number of chiral SM generations; that number is fixed by  $\mathcal{D}_R$ .

## IX. DEGENERACY, KINETICS, VISIBILITY, AND FORMATION

The structures in this section are an effective formation/visibility model built on the  $\text{Spin}^c$  generation theorem. They use scalar spectral data and two-center collective coordinates to parametrize rates and overlaps; they do not supply an independent proof of the number of chiral SM generations. The interface data used below are precisely  $\mathfrak{D}_{\text{int}}(D)$  from Eq. (2); changing them changes the effective formation/visibility model, not the  $\text{Spin}^c$  multiplicity result.

For the protected chiral sector, let  $\Pi_N = |u_N\rangle\langle u_N|$  be the projector onto the Toeplitz eigenmode  $u_N$ . The chiral degeneracy is

$$g_N^{\text{chiral}} = \text{Tr}_{\mathcal{H}_{\text{gen}}} \Pi_N, \quad \mathcal{H}_{\text{gen}} = H^0(\mathbb{CP}^1, \mathcal{O}(2)). \quad (84)$$

Thus

$$\boxed{g_1^{\text{chiral}} = g_2^{\text{chiral}} = g_3^{\text{chiral}} = 1, \quad g_{N \geq 4}^{\text{chiral}} = 0.} \quad (85)$$

### A. Generation-to-spatial mode assignment

The protected generation basis  $u_N \in H^0(C, L_R)$  and the scalar spatial eigenmodes  $\psi_n$  of Eq. (83) are not the same objects. The former are  $\text{Spin}^c$  zero modes on  $C$ ; the latter are eigenfunctions of a Schrödinger operator on the spatial two-center chart. We reserve  $N = 1, 2, 3$  for protected generation labels and  $n$  for scalar spectral labels. Define

$$\mathcal{H}_{\text{gen}} = H^0(C, L_R), \quad \mathcal{H}_{\Omega}(D) = L^2(\mathbb{R}^3, d\mu_{\text{cyl}}), \quad d\mu_{\text{cyl}} = \rho \, d\rho \, dz_{\text{ax}} \, d\varphi, \quad (86)$$

and

$$L_{\Omega}(D) := -\nabla^2 + V_{\text{eff}}(\cdot; D), \quad L_{\Omega}(D)\psi_n(\cdot; D) = \omega_n^2(D)\psi_n(\cdot; D). \quad (87)$$

We use the Friedrichs self-adjoint realization of  $L_{\Omega}(D)$  associated with the semibounded quadratic form of Eq. (87) on  $\mathcal{H}_{\Omega}(D)$ , after the regulated two-center potential  $V_{\text{eff}}(\cdot; D)$  has been fixed. In finite-volume numerical applications,  $L_{\Omega}(D)$  denotes the corresponding self-adjoint realization with the stated boundary conditions. The notation  $dE_{\lambda}(D)$  below is the projection-valued spectral measure of this self-adjoint operator. The projection map  $\pi_D$  pulls the generation bundle to the spatial chart:

$$\hat{L}_R := \pi_D^* L_R, \quad \hat{h}_R := \pi_D^* h_R, \quad \hat{u} := \pi_D^* u \in \Gamma(\mathbb{R}^3, \hat{L}_R). \quad (88)$$

To attach scalar barrier data to a protected generation, introduce the linear raw scalarization operator

$$\boxed{\tilde{\mathcal{C}}_D : \mathcal{H}_{\text{gen}} \longrightarrow \mathcal{H}_{\Omega}(D), \quad \tilde{\mathcal{C}}_D u = w_D^{1/2} \langle \mathfrak{j}_D, \pi_D^* u \rangle_{\hat{h}_R}.} \quad (89)$$

Here  $\mathbf{j}_D \in \Gamma(\mathbb{R}^3, \hat{L}_R)$  is the locking/probe section supplied by the two-center projection data, and  $w_D \geq 0$  is a scalar density weight. With our convention that the Hermitian pairing is antilinear in the first entry and linear in the second,  $\tilde{\mathcal{C}}_D$  is linear in  $u$ . The raw scalar profile is assumed nonzero on the protected modes  $u_N$ . The normalized profile map is the generally nonlinear map

$$c_D(u) = \frac{\tilde{\mathcal{C}}_D u}{\|\tilde{\mathcal{C}}_D u\|_{\mathcal{H}_\Omega(D)}}. \quad (90)$$

The spectral weights of a generation mode against scalar eigenmodes are

$$\mu_{nN}(D) = \left| \langle \psi_n(\cdot; D), c_D(u_N) \rangle_{\mathcal{H}_\Omega(D)} \right|^2. \quad (91)$$

For a normalized discrete scalar spectral resolution,  $\sum_n \mu_{nN} = 1$ . If the scalar operator has a continuous component, Eq. (91) is replaced by the projection-valued spectral measure

$$d\mu_N^D(\lambda) = \|dE_\lambda(D) c_D(u_N)\|_{\mathcal{H}_\Omega(D)}^2. \quad (92)$$

Only when a unique scalar mode dominates do we define a hard mode-locking map. The condition is

$$n(N; D) = \arg \max_n \mu_{nN}(D), \quad \mu_{n(N; D), N}(D) - \sup_{m \neq n(N; D)} \mu_{mN}(D) \geq \Delta_{\text{lock}} > 0. \quad (93)$$

In that regime,

$$\mathcal{I}_D(u_N) := \psi_{n(N; D)}(\cdot; D), \quad \psi_N^{\text{lock}} := \psi_{n(N; D)}, \quad \omega_N(D) := \omega_{n(N; D)}(D). \quad (94)$$

If Eq. (93) fails, the generation must be assigned scalar data softly:

$$\Gamma_N^{\text{soft}}(D, \eta) = \sum_n \mu_{nN}(D) \Gamma_n(D, \eta), \quad (95)$$

or, for a continuous spectral component,

$$\Gamma_N^{\text{soft}}(D, \eta) = \int \Gamma(\lambda; D, \eta) d\mu_N^D(\lambda). \quad (96)$$

Here  $\Gamma_n$  is the scalar-mode formation rate for the  $n$ th eigenmode, computed by the same two-center Schur/WKB construction. Thus the Spin<sup>c</sup> theorem fixes  $u_N$ , while  $\tilde{\mathcal{C}}_D$ ,  $c_D$ ,  $\mu_{nN}$ , and, only under Eq. (93),  $\mathcal{I}_D$ , supply the scalar formation interface. All hard-locking quantities  $\omega_N$ ,  $\Gamma_N$ , and  $|N, \pm\rangle$  below are understood only after Eq. (93); the final probability law is written with  $\Gamma_N^{\text{eff}}$  to cover both the hard-locking and soft-assignment regimes.

For kinetics in the hard-locking regime, the locked scalar mode  $\psi_N^{\text{lock}}$  and the two centers define a collective-coordinate subspace

$$\mathcal{C}_N = \text{span}\{|N, +\rangle, |N, -\rangle\}. \quad (97)$$

Let  $P_N^{\text{cc}}$  project onto  $\mathcal{C}_N$ ,  $Q_N^{\text{cc}} = 1 - P_N^{\text{cc}}$ , and let  $\mathcal{K}_N(D, \eta)$  be the Euclidean fluctuation operator in this sector. The dimensionless parameter  $\eta$  denotes the two-center overlap controlling kinetic mixing. For the regulated source model, we take

$$\boxed{\eta(D, \varepsilon) = \int_{\mathbb{R}^3} d^3x \sqrt{\rho_\varepsilon(x - x_+) \rho_\varepsilon(x - x_-)}, \quad 0 \leq \eta \leq 1.} \quad (98)$$

It enters the fluctuation operator, WKB prefactors, and center mixing coefficient:

$$\mathcal{K}_N = \mathcal{K}_N(D, \eta), \quad A_{N,\pm} = A_{N,\pm}(D, \eta), \quad \chi_N = \chi_N(D, \eta). \quad (99)$$

The Feshbach–Schur step assumes a gap on the eliminated complement:

$$0 \notin \text{Spec}(Q_N^{\text{cc}} \mathcal{K}_N(D, \eta) Q_N^{\text{cc}}). \quad (100)$$

If this gap closes, the inverse below must be replaced by the corresponding reduced resolvent or pseudoinverse. Under Eq. (100), the Feshbach–Schur effective matrix is

$$\mathcal{M}_N = P_N^{\text{cc}} \mathcal{K}_N P_N^{\text{cc}} - P_N^{\text{cc}} \mathcal{K}_N Q_N^{\text{cc}} (Q_N^{\text{cc}} \mathcal{K}_N Q_N^{\text{cc}})^{-1} Q_N^{\text{cc}} \mathcal{K}_N P_N^{\text{cc}}. \quad (101)$$

In the two-center basis,

$$\mathcal{M}_N = \begin{pmatrix} \gamma_{N,+} & \chi_N \sqrt{\gamma_{N,+} \gamma_{N,-}} \\ \chi_N \sqrt{\gamma_{N,+} \gamma_{N,-}} & \gamma_{N,-} \end{pmatrix}, \quad (102)$$

with WKB channel factors

$$\gamma_{N,\pm} = A_{N,\pm} \exp[-2S_{N,\pm}^{\text{WKB}}], \quad S_{N,\pm}^{\text{WKB}} = \int_{\mathcal{B}_{N,\pm}} ds \sqrt{(V_{\text{eff}}(s) - \omega_N^2)_+}. \quad (103)$$

Here

$$\mathcal{B}_{N,\pm} = \{s \text{ on the } \pm \text{ channel} : V_{\text{eff}}(s) > \omega_N^2\}, \quad (104)$$

and  $(x)_+ = \max\{x, 0\}$ . The formation rate is

$$\boxed{\Gamma_N(D, \eta) = \max\{0, \lambda_+(\mathcal{M}_N(D, \eta))\}}, \quad (105)$$

where

$$\lambda_+(\mathcal{M}_N) = \frac{\gamma_{N,+} + \gamma_{N,-}}{2} + \frac{1}{2} \sqrt{(\gamma_{N,+} - \gamma_{N,-})^2 + 4\chi_N^2 \gamma_{N,+} \gamma_{N,-}}. \quad (106)$$

For visibility, use the same pullback bundle from Eq. (88). For the  $N$ th generation, the spatial representative is the section

$$\hat{u}_N := \pi_D^* u_N \in \Gamma(\mathbb{R}^3, \hat{L}_R), \quad (107)$$



not a scalar function. The visible kernel is correspondingly a bundle-valued source

$$\mathcal{J}_{\text{vis}}(D, \varepsilon) \in \Gamma(\mathbb{R}^3, \hat{L}_R). \quad (108)$$

The  $\varphi$ -dependence is therefore not discarded; in axially reduced formulas it has already been projected into the selected angular channel and absorbed into the chosen representative of  $\mathcal{J}_{\text{vis}}$  and the measure factor. The conformal Dirac and Higgs fields scale as

$$\Psi = \Omega^{-3/2} \psi, \quad H = \Omega^{-1} \phi. \quad (109)$$

Hence

$$\sqrt{-g} \bar{\Psi} H \Psi \propto \Omega^4 \cdot \Omega^{-3/2} \cdot \Omega^{-1} \cdot \Omega^{-3/2} = \Omega^0. \quad (110)$$

A mode-resolved Yukawa overlap is the Hermitian pairing on the pullback bundle:

$$y_N^{\text{eff}}(D, \varepsilon) = y_0 \int_{\mathbb{R}^3} \langle \hat{u}_N(x; D, \varepsilon), \mathcal{J}_{\text{vis}}(x; D, \varepsilon) \rangle_{\hat{h}_R} d\mu_{\text{cyl}}(x), \quad (111)$$

We use the convention that the Hermitian pairing is antilinear in the first entry. With this mathematical convention,  $y_N^{\text{eff}}$  is an antilinear amplitude in the generation vector  $u_N$ . This does not affect  $B_N$ , which depends only on  $|y_N^{\text{eff}}|^2$ . If  $y_N^{\text{eff}}$  is later used as a phase-sensitive Yukawa or Wilson coefficient and one wants a convention linear in  $u_N$ , one may equivalently use the opposite pairing order,  $\int \langle \mathcal{J}_{\text{vis}}, \hat{u}_N \rangle_{\hat{h}_R} d\mu_{\text{cyl}}$ ; the normalized visibility  $B_N$  is unchanged. In a local frame  $\hat{e}_R = \pi_D^* e_R$ , write

$$\hat{u}_N = U_N \hat{e}_R, \quad \mathcal{J}_{\text{vis}} = J_{\text{vis}} \hat{e}_R, \quad \hat{h}_R(\hat{e}_R, \hat{e}_R) = H_R. \quad (112)$$

Then

$$\langle \hat{u}_N, \mathcal{J}_{\text{vis}} \rangle_{\hat{h}_R} = H_R \overline{U_N} J_{\text{vis}}. \quad (113)$$

This is independent of the frame. Indeed, under  $\hat{e}'_R = g \hat{e}_R$ , with  $g : \mathbb{R}^3 \rightarrow \mathbb{C}^*$ ,

$$U'_N = g^{-1} U_N, \quad J'_{\text{vis}} = g^{-1} J_{\text{vis}}, \quad H'_R = |g|^2 H_R, \quad (114)$$

and hence

$$H'_R \overline{U'_N} J'_{\text{vis}} = H_R \overline{U_N} J_{\text{vis}}. \quad (115)$$

The earlier scalar-looking formula is only this local trivialized representative:

$$y_N^{\text{eff}}(D, \varepsilon) = y_0 \int_0^\infty \rho d\rho \int_{-\infty}^\infty dz_{\text{ax}} \int_0^{2\pi} d\varphi \overline{U_N(\rho, z_{\text{ax}}, \varphi; D, \varepsilon)} \\ \times f_L(\rho, z_{\text{ax}}) f_R(\rho, z_{\text{ax}}) W_{\text{frame}}(\rho, z_{\text{ax}}, \varphi; D, \varepsilon). \quad (116)$$

Here  $W_{\text{frame}}$  is not an additional invariant object; it denotes the local representative of the bundle frame, Hermitian metric  $H_R$ , visible source  $J_{\text{vis}}$ , angular-channel projection, Jacobian convention, and any remaining complex-conjugation convention. When the visible kernel has already been reduced to an  $m = 0$  axial channel, Eq. (116) reduces to the earlier  $2\pi\rho d\rho dz_{\text{ax}}$  form with the angular integral included in  $W_{\text{frame}}$ . The normalized visible weight is

$$B_N = \frac{|y_N^{\text{eff}}|^2}{\sum_{I=1}^3 |y_I^{\text{eff}}|^2}. \quad (117)$$

For the operator notation in the equivalent form, define the linear visible-overlap functional

$$\mathcal{O}_{\text{vis}} : \mathcal{H}_{\text{gen}} \longrightarrow \mathbb{C}, \quad (\mathcal{O}_{\text{vis}} u) = y_0 \int_{\mathbb{R}^3} \langle \mathcal{J}_{\text{vis}}(x; D, \varepsilon), \pi_D^* u(x) \rangle_{\hat{h}_R} d\mu_{\text{cyl}}(x). \quad (118)$$

This uses the opposite pairing order from Eq. (111), so it is linear in  $u$  under the stated Hermitian convention. The two overlap conventions differ only by pairing order and conjugation convention and give

$$|\mathcal{O}_{\text{vis}} u_N|^2 = |y_N^{\text{eff}}|^2. \quad (119)$$

Thus  $\mathcal{O}_{\text{vis}}^\dagger \mathcal{O}_{\text{vis}}$  is the induced rank-one positive operator on  $\mathcal{H}_{\text{gen}}$ , and equivalently

$$B_N = \frac{\langle u_N | \mathcal{O}_{\text{vis}}^\dagger \mathcal{O}_{\text{vis}} | u_N \rangle}{\sum_{I=1}^3 \langle u_I | \mathcal{O}_{\text{vis}}^\dagger \mathcal{O}_{\text{vis}} | u_I \rangle}. \quad (120)$$

Finally, at the coarse-grained effective level, define the rate used by the conditional probability law as

$$\Gamma_N^{\text{eff}}(D, \eta) = \begin{cases} \Gamma_N(D, \eta), & \text{in the hard-locking regime of Eq. (93),} \\ \Gamma_N^{\text{soft}}(D, \eta), & \text{otherwise.} \end{cases} \quad (121)$$

Assume the corresponding single-rate Markovian formation law

$$\frac{dp_N}{dt} = \Gamma_N^{\text{eff}}(1 - p_N), \quad p_N(0) = 0 \quad (122)$$

has solution

$$p_N(t) = 1 - e^{-\Gamma_N^{\text{eff}} t}. \quad (123)$$

Conditioning on the observation of one of the protected chiral layers gives

$$P_N(t; D, \eta) = \frac{B_N g_N^{\text{chiral}} \left(1 - e^{-\Gamma_N^{\text{eff}}(D, \eta) t}\right)}{\sum_{K=1}^3 B_K g_K^{\text{chiral}} \left(1 - e^{-\Gamma_K^{\text{eff}}(D, \eta) t}\right)}. \quad (124)$$

## X. CONCLUSION

The two-center PSLT projection data determine the divisor  $R = p_+ + p_-$  and the line bundle  $L_R = \mathcal{O}(R) = \mathcal{O}(2)$  on  $C = \mathbb{CP}^1$ . The SM-charged internal fermions are governed by the product Dirac action on  $M_4 \times C$ , with the internal  $\text{Spin}^c$  Dolbeault–Dirac operator acting on  $\Lambda^{0,*}T^*C \otimes L_R$ . The bundle  $L_R$  is the auxiliary Dolbeault twist, and the positive and negative internal zero modes are  $H^0(C, L_R)$  and  $H^1(C, L_R)$ . Riemann–Roch gives index three, and Serre duality gives  $H^1(C, L_R) = 0$ . With all additional SM-charged sectors either absent from the kernel or fully paired by a gauge-invariant mass map, the theory therefore has exactly three low-energy chiral SM generations and no unpaired light SM-charged exotics.

After the raw scalarization operator  $\tilde{\mathcal{C}}_D$ , the normalized profile map  $c_D$ , the spectral weights  $\mu_{nN}$ , and, when the locking gap exists, the hard map  $\mathcal{I}_D : u_N \mapsto \psi_N^{\text{lock}}$ , the scalar two-center spectral operator and the quantities  $(g_N^{\text{chiral}}, \Gamma_N^{\text{eff}}, B_N, P_N)$  define an effective model for how these three protected modes form and couple to the visible sector. They are coarse-grained phenomenological structures on top of the  $\text{Spin}^c$  generation theorem, not substitutes for it.

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